

3(a)	one similarity ; e.g. substrate binds to active site enzyme-substrate complex forms product leaves active site interaction does not alter enzyme for re-use lower activation energy specific to one or very similar substrate(s) interaction can be affected by, action of inhibitors / pH / AW one difference ; e.g. induced-fit ora for lock-and-key active site not fully complementary to substrate change in, conformation / shape, of active site / AW <i>context of binding substrate or after product release</i> less specificity to substrate	2
3(b)(i)	orange / brown / amber ; A yellow-brown, yellow-orange	1
3(b)(ii)	any three from: phosphorylase (in extract) catalyses addition of glucose 1-phosphate to starch added to extract ; more starch is synthesised by phosphorylase as time progresses / AW ; iodine solution added to mixture turns a, blue / black / blue-black, colour (in the presence of starch) ; the more starch present in the mixture, the, darker / more intense, the, colour / shade ; the darker the colour (of the positive result) the less light passes through, mixture / sample ; the darker the colour / AW, the higher the absorbance (reading of the colorimeter) ;	3
3(b)(iii)	<i>ref. to</i> breakdown of starch catalysed by the enzyme / reaction in Fig. 3.1 goes in the opposite direction ; AVP ; e.g. <i>ref to</i> equilibrium shift <i>ref. to</i> glucose 1-phosphate has run out decrease in concentration of starch, qualified with <i>ref. to</i> iodine solution	1

Question	Answer	Marks
3(c)	<i>any three from:</i> reduction in activity of phosphorylase (at all concentrations) ; ora higher throughout with caffeine $V_{\max}$ not reached ; <i>ref. to</i> plateau at a (much) lower activity ; calculated K_m value with caffeine is the same as without caffeine ; use of data from the graph to support conclusion ;	3

Question	Answer	Marks
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